1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: An integrated multi-modal prognostic score (MTPS) that combines image-derived histopathology features (NE_frac, Cellularity, Nuclear_Grade_Het, Gland_Disruption, Stroma_Ratio) with molecular scores (AR_Score, CCP_Score, NE_Score, and DRD/EMT-related features) will provide superior prognostic stratification for overall survival (OS) and time-to-progression (TTP) in patients with metastatic/advanced disease, compared with standard clinicopathologic factors or any single modality alone.

Secondary hypotheses: - H1: The MTPS will achieve higher discrimination (e.g., C-index, time-dependent AUC) for OS and TTP than a model using only clinicopathologic variables or only image-derived features, across multiple independent cohorts and platforms. - H2: MTPS-defined risk groups (low/medium/high) will show distinct Kaplan-Meier survival curves with clinically meaningful separation after adjusting for confounders. - H3: The MTPS is robust to cross-site and platform heterogeneity (e.g., different RNA platforms and imaging workflows) when harmonization steps are applied, with minimal loss of performance in external validation cohorts. - H4: Component analysis will reveal that AR_Score and CCP_Score contribute most to discriminative power in high-risk groups, while NE_Score and EMT-related features modulate risk in different metastatic microenvironments (e.g., visceral vs. bone). - H5: A minimal clinically feasible subset of features (e.g., a core set of 4–6 features) can reproduce most of the MTPS performance, enabling near-term clinical translation while maintaining calibration.

Aims and testing strategy: - Aim 1: Develop a robust MTPS by integrating standardized, literature-backed histopathology features with validated RNA-derived scores; assess prognostic performance for OS and TTP in retrospective multi-center cohorts. - Aim 2: Validate MTPS in external cohorts with diverse platforms (bulk RNA-seq, 3′-seq, targeted panels) and imaging modalities; evaluate cross-platform harmonization strategies. - Aim 3: Conduct interpretable model analyses to quantify feature contributions and identify a parsimonious feature set suitable for prospective translation; define clinically actionable risk thresholds.

Experimental approach (high-level): - Data collection: Assemble retrospective cohorts with matched histology images, NE IHC data when available, RNA expression data (preferred platforms include bulk RNA-seq or equivalent), and clinical outcomes (OS, TTP). Record site, platform, and assay details for harmonization. - Feature extraction: Compute image-derived metrics (NE_frac, Cellularity, Nuclear_Grade_Het, Gland_Disruption, Stroma_Ratio) via validated AI/pathology pipelines or standardized pathologist scoring; compute molecular scores (AR_Score, CCP_Score, NE_Score) and EMT-related features from RNA data using established gene-sets; include DRD-related markers as exploratory features. - Data preprocessing: Implement robust missing-data strategies, platform normalization, and cross-site batch effect correction (e.g., ComBat or hierarchical models). - Modeling: Build an elastic-net Cox proportional hazards model and/or joint modeling framework to integrate image and molecular features; use nested cross-validation with bootstrapping to estimate performance (C-index, time-dependent AUC), calibration plots, and decision-curve analyses. - Validation: Test on independent external cohorts; perform sensitivity analyses by excluding sites, altering lesion- vs patient-level aggregation rules, and evaluating alternative endpoints (e.g., therapy-specific failure). - Translation: Derive a clinically actionable risk threshold scheme and provide a minimal feature subset with validated performance for potential CLIA-ready workflows.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions